

CLASS-12

SENIOR SECONDARY EXAMINATION -2017

BIOLOGY

TIME: 3 ¼ HOURS

MM: 56

GENERAL INSTRUCTION TO THE EXAMINEES.

1. Candidates must write first his/ her roll number on the question paper compulsorily.
2. All questions are Compulsory.
3. Write the answer to each question in the given answer book only.
4. For questions having more than one part the answers to those parts are to be written together in continuity.
5. If there is any error difference contradiction in Hindi and English versions of the question paper the questions of Hindi version should be treated valid.
6. For questions having more than one part the answers to the other part should be written together in continuity.

1. Section	q.no	marks
A	1-13	1
B	14-24	2
C	25-27	3
D	28-30	4

2. There are internal choices in question 24, 27, 28, 29 and 30.

SECTION-A

- 1) **What are morphologically and genetically alike individuals called? [1]**
Ans: Morphological and genetically alike individuals are called clones .The process of development of embryo of zygote is called embryogenesis
- 2) **Defined embryogenesis. [1]**
Ans: The process of development of embryo of zygote is called embryogenesis
- 3) **Write the importance of colostrum. [1]**

Ans: The milk produced during the initial few days of lactation is called colostrum which contains several antibodies absolutely essential to develop resistance for new born baby. Colostrum also helps the newborn digestive system to grow and function properly.

- 4) **Write the full form of IVF.** [1]

Ans: In vitro fertilization is the full form.

- 5) **Name anyone intrauterine device.**

Ans: Example of intrauterine device copperT, cuT 380, multiload 375.

- 6) **What is genotype of a person having kline filter syndrome ?[1]**

Ans: Genotype of a male having kline filter syndrome is XXX with 47 chromosome.

- 7) **Genetic code is unambiguous and specific. Explain.**
[1]

- 8) **Who propounded the biological evolution theory of use and disuse of organs?**

Ans: Jean Baptist De lamarck propounded the biological evolution theory of use and disuse of organs.

- 9) **Mention the importance of Blue revolution.** [1]

Ans: It result in increase in production of fish.

- 10) **How do ultraviolet rays coming from sun affect the living organism?**
[1]

Ans: It may cause skin cancer, risk in ageing and mutation and eye cataract in organism.

- 11) **What are ill effects of algal blooms?** [1]

Ans: Algal blooms imparts a distinct colour to the water bodies. Algal Bloom causes deterioration of the water quality and fish mortality. Some Bloom forming Algae are extremely toxic to human beings and animals.

- 12) **Define Recombinant DNA.** [1]

Ans: Recombinant DNA is formed by linking a foreign gene with plastid and is then transferred into the host so that DNA replicates and benefits of orange in can be obtained.

- 13) **The Bt toxin produced by Bt does not kill Bacillus but insects .why?** [1]

Ans: BT toxin is active in insect because of alkaline pH of the gut of insect and causes death of the insect but it is inactive in bacillus.

SECTION-B

14) **What is spermiogenesis? Demonstrate diagrammatically the process of spermatogenesis. [1+1=2]**

Ans: The process of transformation of a circular spermatid to spermatozoa is called Spermiogenesis. In this process the nucleus of the spermatid becomes the head of the spermatozoa. The Golgi apparatus of the spermatid containing proteolytic enzyme becomes the acrosomal cap.

15) **What is transcription? What are different parts of DNA transcription unit? What is their role in process of transcription? [$\frac{1}{2} + \frac{1}{2} + 1 = 2$]**

Ans: The first process in the overall information transfer from DNA into protein structure is called transcription. During this process a complementarily messenger RNA strand is formed from DNA strand. Transcription unit in DNA is defined primarily by three regions in the DNA.

- i) Promoter.
- ii) Structural gene.
- iii) Terminator

The promoter and Terminator flank the structural gene in a transcription. Terminator usually define the end of process of transcription.

16) **With the help of examples explain the process of biological evolution on the basis of morphological evidences. [2]**

Ans: Comparative morphology and anatomy of individual organ and organ system of diverse form reveal their phylogenetic relationship and thus provides important evidence of the evolution.

Organs different in function but of similar embryo genic origin and development have similar relationships with adjacent organ are known as homologous organ for example-arms of man, leg of horse, wing of a bird or a bat, flipper of a seal and forelimb of a cheetah.

Analogy and analogous organs -Analogous organs are those which perform the same function has superficial resemblance to one another but are of different origin and this phenomena is called analogy for example Wings of insects ,Birds and bats

17) **.Write down in sequence the main steps of plant breeding .What is its importance? [2]**

Ans: Plant breeding is the purposeful many manipulation of plant species in order to crease desired plant types that are better suited for cultivation give better yield and disease resistant. Main steps of plant are following-

- Collection of variability.

- Evaluation and selection of different
- Hybridization among the selected parents.
- Selection and testing of superior Recombinant
- Testing release and commercialisation of new cultivar.

18) What suggestion will you give a farmer to increase the productivity of his field?

Explain [2]

Ans: Following are the methods which can be suggested to farmer's farmer to increase the productivity of his field-

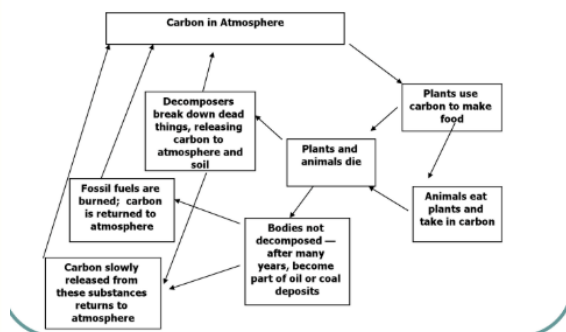
- Plant breeding is a specialized Technology based on genetics. The main aim of plant breeder is to synthesise a genetically superior quality plant which is adaptable to a specific set of environment.
- To provide sufficiently fast and efficient system of Crop improvement tissue culture technique has been developed.
- Single cell protein is the alternative sources of protein for animal and human nutrition.

19) With reference to fermented beverage explain the role of microbes in Industrial Products.

Ans: Microorganism possess several characteristics which make them ideal organism for industrial processes. They contain a broad variety of enzymes making an array of chemical conversion possible. Microbes specially yeast are used for production of beverages like wine, beer, whisky, brandy or rum. Depending on the type of raw materials used for fermentation and the type of processing different types of alcoholic drinks are obtained. Wine and beer are produced without distillation where whisky brandy and rum are produced distillation of fermented food.

19 Explain carbon cycle in ecosystem .give a linear diagram of carbon cycle.

Ans: Carbon constitutes of 49% of dry weight of organism. 4 into 10^{13} kg of carbon is fixed in the biosphere by photosynthesis. Large amount of carbon return to atmosphere as carbon dioxide through respiration of Producers and Consumers. Decomposers also return carbon dioxide to reservoir during the composition process. Some amount of carbon is lost to sediments and are removed from circulation. Burning wood, forest fire, emission of organic matter, fossil fuel also release carbon dioxide to atmosphere. Rapid deforestation and Massive burning of fossil fuels for energy and transport have significantly increased the rate of release of carbon dioxide into atmosphere.



CARBON CYCLE

20 What is adaptation ? Explain physiological adaptation taking an example of altitude illness .[$\frac{1}{2} + 1 \frac{1}{2} = 2$]

Ans: Adaptation is any attribute of the organism which may be morphological physiological behavior behavioral that enables the organism to survive and reproduce in its habitat.

Physiological adaptation- At high altitude places we can experience altitude sickness. Its symptoms include nausea, heart palpitation, fatigue .This is because in low atmospheric pressure of altitude body does not get enough oxygen but gradually body gets acclimatised and stop experiencing at altitude sickness .The body compensate the low oxygen availability by increasing red blood cell production, decreasing the binding capacity of hemoglobin and increasing breathing rate.

21 Give four attributes of a population .Explain in detail the process of mutualism. [1+1=2]

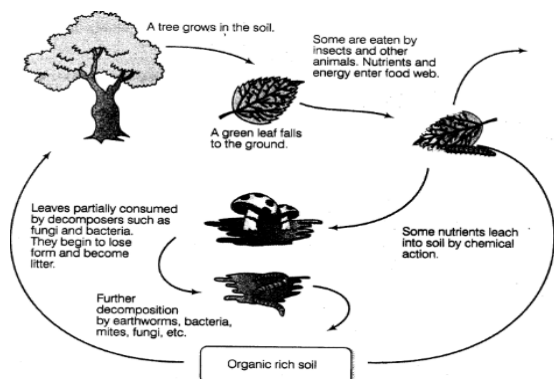
Ans: Four attributes of population are following-

- Population Density
- Birth Rate
- Death Rate
- Sex Ratio

Mutualism is an interaction between two species in which both are benefited .It maybe symbiotic in which organisms live together in close Association. Lichens, mycorrhiza , symbiotic nitrogen fixers are well-known examples of Symbiosis.

22 What is decomposition? Write down its steps. Give a diagrammatic representation of decomposition cycle in Terrestrial ecosystem. [$\frac{1}{2} + \frac{1}{2} + 1 = 2$]

Ans: Decomposition is a process whereby decomposers break down Complex organic matter into inorganic substances like carbon dioxide and water and nutrients. The main steps in the process of decomposition are fragmentation, leaching, catabolism, humification and mineralization.



Decomposition cycle in Terrestrial ecosystem.

23 Give four causes of loss of biodiversity. Explain the adverse effects on native fishes on invasion of alien species in a habitat.

[1+1=2]

Some important causes of biodiversity losses are following –

- Habitat loss and fragmentation
- Over exploitation
- Alien species invasion
- Pet Trade
- Co- extinction

When alien species are introduced unintentionally, some of them turn invasive and causes the decline or extinction of indigenous species. The Nile Perch introduced into Lake Victoria in East Africa led it eventually to the extinction of more than 200 species of cichlid fish in the lake.

OR

How is conservation of biodiversity done? How will we conserve an animal to save it from extinction?

Ans: Conservation of biodiversity is done in following ways-

In situ conservation means maintaining plant diversity in natural reserves. It protects the species threatened with loss of extinction, supply material to restore degraded land and provide material for genetic improvement of crop plants. It is mostly done by establishing biosphere reserve, national park, and wildlife sanctuaries.

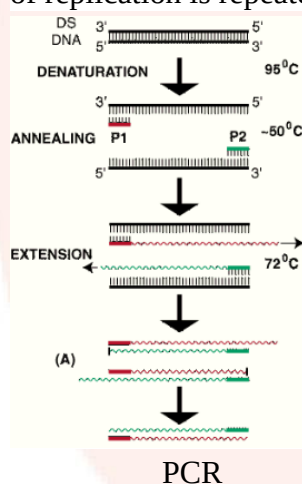
Ex situ conservation-The cultivation of selected rare plants in Botanical Garden constitutes ex situ conservation. Besides the primary repositories of wild plants, botanical gardens have a crucial role in conservation and education. As some plants like ginkgo tree have become so common

simple because this multiplication was taken up by gardeners. Ex situ conservation of animal genetic resources includes 'cryogenic preservation techniques'. The collection and freezing in liquid nitrogen of semen, ova or embryo, or the preservation of DNA segment in frozen blood or other tissues.

SECTION –C

24) How is the amplification of change done using the technique of PCR. Explain with the help of diagram. [2+1=3]

Amplification of gene of interest using PCR-In this reaction multiple copies of genes or DNA of interest is synthesized in vitro using two sets of primers and the enzyme DNA polymerase. The enzyme extends the primers using the nucleotides provided in the reaction and the genomic DNA as a template. The repeated amplification is achieved by use of thermo stable DNA polymerase which is isolated from bacterium *thermus aquaticus*. It remain active during the high temperature induced denaturation of double stranded DNA. The amplified fragment can then be used to ligate it with vector for further cloning. Each cycle has three steps -,denaturation primer annealing and extension of primers. If the process of replication is repeated many times 1 billion copies of desired DNA obtained.



25) What is the importance of biotechnology application in medicine? How does the biotechnology help in the production of human insulin? [1+2=3]

Ans: The Recombinant DNA technology processes made immense impact in the area of Healthcare by enabling mass production of safe and more effective therapeutic drugs. Improve Diagnostic tools are developed, Immunological therapeutic methods are used and production of novel medicines vaccines are some of the benefits of biotechnology. It is possible to obtain vitamins on a large scale by the use of microorganisms. New vaccines known as second generation vaccines and third generation vaccines have been developed. An American firm Eli Lilly launched a product known as humulin by successful transfer of human insulin gene into bacteria *E.coli*. Two DNA sequences corresponding to chains A and

B, Chains of human insulin were first prepared and introduced into the plasmids pBR 322 of E. Coli to produce insulin chains. Both chain were produced separately, extracted and combined by creating disulfide bonds to form human insulin. Insulin was the first genetically engineered protein to be used in humans and it is now used in routine clinical practice throughout the world.

26) DNA is a genetic material. Prove it with the help of Hershey chase experiment.

Draw a linear diagram of the experiment.

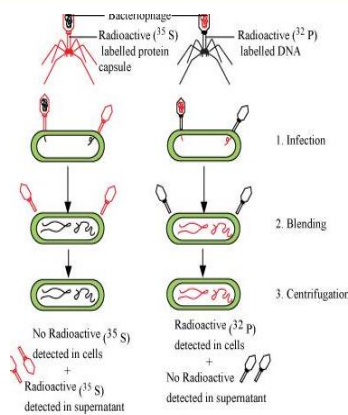
Ans: Alfred De Hershey and Chase conducted experiments on bacteriophages that attack bacteria. They concentrated on T2 bacteriophages that attack a common bacteria-Escheria Coli in human digestive tract. This bacteriophage has only two components protein and DNA. They realise that this bacteriophage bacterium is ideal material for solving the DNA versus protein controversy.

Experiment –

They separated labeled the protein and DNA components of bacteriophages with specific radioactive tracers and then followed this component through phages life cycle. They develop two strengths of viruses one with labelled protein and other with labelled DNA. Almost all protein contains Sulphur and atom not found in DNA whereas all DNA molecules contain Phosphorus and atom not found in protein. Thus bacteriophages parasitizing bacteria grown in the presence of radioactive sulphur (^{35}S) had labelled protein and bacteriophages parasitizing bacteria grown in the presence of radioactive Phosphorus (^{32}P) labelled DNA. Such differential labelling thus enables to distinguish between DNA and protein of the phage. After developing this to strain Hershey and Chase combined each strain with non-radioactive bacteria and allowed bacteriophages to attach and inject their genetic material. Soon after infection the bacterial cell were gently agitated in a Blender to loosen the adhering phage particle.

Result- it was observed that only reactive Radioactive (^{32}P) was found associated with bacterial cells and (^{35}S) was only in surrounding medium and not in the bacterial cells. When phage progeny was studied For radioactivity it was found that it carried label only with (^{32}P) and not with ^{35}S . This result clearly indicate that only DNA and not protein coat enter the bacterial cell.

Conclusion-It can be concluded that it is a viral DNA and not the protein that contain information for production of mode viral particle. Hershey and Chase have shown that DNA not protein carries hereditary information.



HERSHEY AND CHASE EXPERIMENT

OR

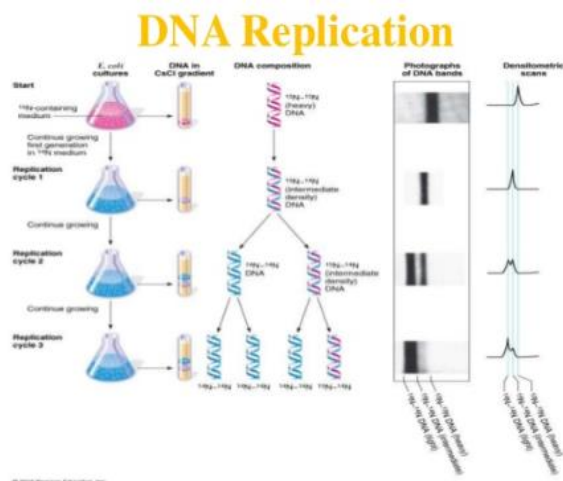
“DNA replicates semi conservatively”. Prove it with the help of Mathew Meselson and Franklin stall experiment draw the linear diagram of the experiment.

Matthew S. Meselson and Franklin W. stall did some experiment to confirm semi conservative nature of DNA.

In one experiment they grew two cultures of E-Coli bacteria for many generations in separate media. The heavy culture was grown in the medium in which nitrogen source ammonium chloride containing the isotopes ^{15}N and the light culture was going in a medium in which nitrogen source was the most common light isotope ^{14}N . At the end of the growth they can observe that bacterial DNA in the heavy culture contains only ^{15}N and light culture contains only ^{14}N . They analyse the two types of bacterial DNA with a technique for caesium chloride density gradient centrifugation. In this process heavy and light DNA extracted from cells in the two cultures were settled into two distinct and separate bands.

In another experiments they transferred heavy culture containing ^{15}N into the medium that held only ^{14}N and leaving the bacteria in light medium long enough to allow two replication of DNA. After 1st replication they extracted DNA and subjected to density gradient centrifugation. The DNA settled into a band that was intermediate in position between the previously determining heavy and light bands. After the second replication, they again extracted DNA samples and this time they found the DNA setting into two bands one at the light band position and the other at the intermediate position.

Conclusion- After the first application there were only intermediate molecules containing both ^{14}N and ^{15}N . This could happen either with semi conservative or dispersive replication but not with conservative replication. This result confirms the semi-conservative replication of DNA.

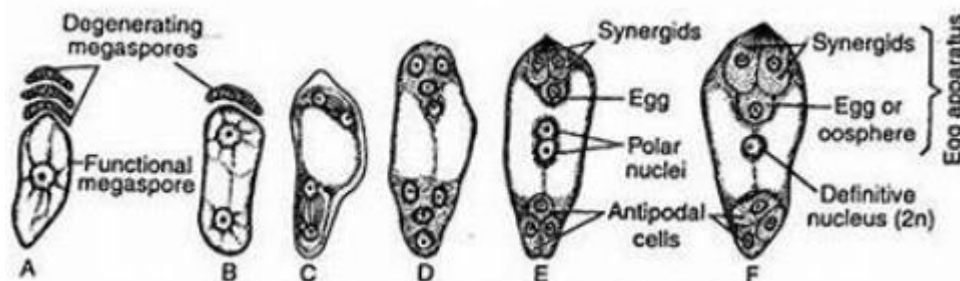


Meselson and Stahl Experiment

SECTION-D

29) Explain in detail the process of development of female gametophyte .Draw diagram [3 + 1=4]

Ans: One Megaspore cell in the linear tetrad towards the chalaza end is active and it forms female gametophyte .Rest of the three megaspore degenerate and the active megaspore uses its nutrients .the active mrgaspore grow and develop into embryo sac which is female gametophyte. The active haploid megaspore is the first cell of female gametophyte. It is also called as embryo sac mother cell .It takes its nutrition from the nucellus and grow bigger and occupies most of the space that develop several small vacuoles which combined to form a large vacuole .The nucleus of megaspore divide mitotically three times to form eight nuclei. The nuclei formed from the first nuclear fusion reach to the opposite poles towards the chalaza and micropyle. Here the nuclei divide two times to produce four nuclei at each pole. From this four nuclei one nucleus in each pole moves towards the center and the form polar nucleus. Later this combined to form diploid secondary nucleus. It is called definitive nucleus. At the edge of micropylar end three nuclei from egg Apparatus. At this stage they get converted into cells by cell wall formation. in the egg Apparatus there are one egg cell and two synergid cells. Similarly at chalazal.end three nuclei combine to form three cells called antipodal cells. The embryo sac of female gametophyte developed by polygonum type has several seven cells and eight nuclei. The secondary nucleus that becomes secondary cell is diploid and the rest cells are haploid.

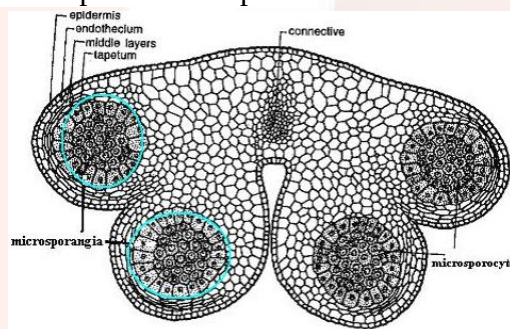


DEVELOPMENT OF FEMALE GAMETOPHYTE

OR

Explain the structure of microsporangium and write the functions of its different layers. Draw the diagram of microsporangium showing wall layers.

Ans: An anther has two lobes. Each lobe has characteristically two microsporangia. A large number of microspores or pollen grain are present in microsporangium. Young anther is usually four chambered at an early stage of development. Some hypodermal cells known archesporium forming a primary parietal cells and a primary sporogenous cell toward the inside. The primary parietal cells divide again form the wall of anther. The wall of the anther is made up of epidermis and endothecium, one or more middle layer and usually a single layer of the Tapetum. The epidermis comprises of tangentially stretched cell. The endothecium has columnar cells. The cells of middle layer are ephemeral and degenerative before the Pollen mother cells undergo meiosis. The tapetum which encases sporogenous tissue has dense cytoplasm cell which maybe multinucleate. The primary sporogenous tissue may either directly or after dividing mitotically function as microspore mother cell and undergo meiosis to form four haploid microspores.



MICROSPORANGIUM

29) Describe AIDS disease on the basis of following points- $\frac{1}{2} + \frac{1}{2} + 1 + 1 + 1 = 4$

- Name of pathogen.
- Test for confirmation of disease.
- Main symptoms of disease.
- Way of prevention.
- Diagram of replication of retrovirus.

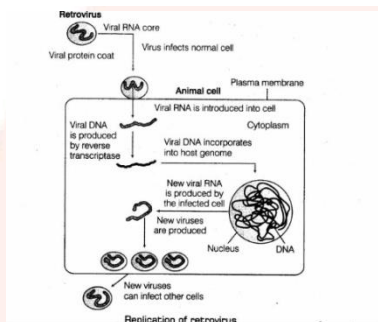
Ans: i) Name of pathogen - AIDS is caused by human immunodeficiency virus (HIV).

ii) Test for confirmation of disease- Diagnostic test for AIDS is enzyme linked immune-sorbent essay (ELISA)

iii) Main symptoms of disease-Fever, diarrhoea .weight loss, hair loss, anaemia

iv) Way of prevention-

- Avoidance of drug abuse.
- Avoiding unsafe human sexual intercourse.
- Blood Door should be free of HIV



OR

Described typhoid disease on the basis of following points-

- Name of pathogen.**
- Test for confirmation of disease.**
- Ways of infection.**
- Min symptoms of the disease.**
- Diagram showing structure of antibody molecule.**

Ans: i) Name of pathogen- Typhoid is caused by bacterium Salmonella typhi.

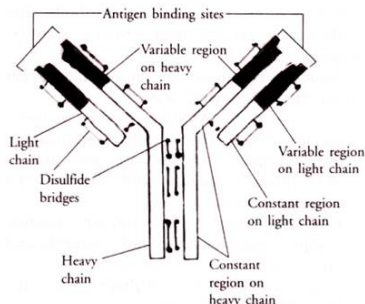
ii) Test for confirmation of disease -Typhoid fever could be confirmed by Widal test.

iii) Ways of infection-Through contaminated food and water.

iv) Major symptoms of typhoid-

- Sustained high fever
- Headache
- Mental confusion.
- Abdominal pain

- The rash of Rose colour spots beginning in the second week of the disease.
- Weakness
- loss of appetite
- Intestinal perforation



Structure of Antibody

30) Explain Mendel's monohybrid hybridization experiment. Write the rules proposed on the basis of this experiment .Draw its diagram using Punnett square. [2+1+1=4]

Ans: For monohybrid cross Mendel begin with a pair of pea plants with two contrasting traits that is one tall and another dwarf. The cross pollination of tall and dwarf plants resulted in tall plants. All the hybrid plants were tall. He called this as a first hybrid generation F₁ and offspring were called filial F₁ for F₁ progeny.

Let T be used for the tall trait and t for the dwarf and T and t are alleles of each other. Hence in plants the pair of Alleles for height would be TT tt or Tt.

TT and tt are genotype of the plant while tall and dwarf are phenotype.

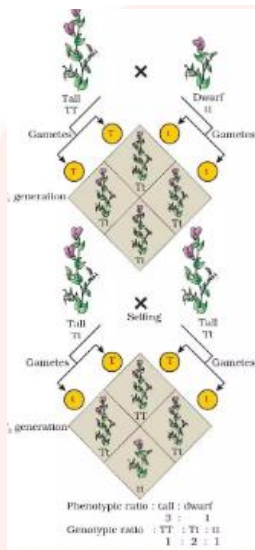
As Mendel found the phenotype of Tt would be exactly like TT parents in appearance he propose that in a pair of dissimilar factor one dominates and the other and hence is called dominant factor while the other factor is recessive. In this case T for tallness is dominant over (dwarfness), that is recessive.

Since the Tt plant is heterozygous for genes controlling one character (height), it is a monohybrid. The cross between two monohybrid traits (TT and tt) is called monohybrid cross. The cross-pollination of tall and dwarf plants resulted in tall plants. All the hybrid plants were tall. He called this as a first hybrid generation (F₁) and offspring were called Filial₁ or F₁ progeny. He conducted an experiment with all the seven contrasting pairs. He observed that the entire F₁ progeny showed one pattern in their behavior i.e., they resembled either one of the parents. Another parent character was completely absent. He continued his experiment with self-pollination of F₁ progeny plants. Surprisingly, he observed that one out of four plants were dwarf while other three were tall. The tall and the short plants were in the ratio of 3:1. He also noted that no progeny was in intermediate height i.e., no blending. The result was same for other traits of plants too. And he called them second hybrid generation and offspring were called Filial₂ or F₂ progeny.

Mendel observed that traits were absent in F₁ generation had reappeared in F₂ generation. He called such suppressed traits as recessive traits and expressed traits as dominant traits. He also concluded that some 'factors' are inherited by offspring from their parent over successive generations.

Rules proposed on the basis Mendel's mono hybridization experiment are also named as the Principles or Laws of Inheritance. These are:

1. Law of dominance-This law states that each character is controlled by distinct units called factors which occurs pairs. If the pairs are heterozygous one will always dominate the other.



PUNNEET SQUARE

2. Law of segregation-This law states that the pair of elements segregates from each other during meiosis cell division so that only one allele will be present in each gamete.

OR

What is co-dominance? Explain its determination of blood groups in human. Draw a table showing the genetic basis of blood groups in human population.

Ans: Co-dominance occurs when two alleles of the same gene are present in a living thing and both are expressed. Instead of one trait being dominant Over the Other both traits appear.

Example of co-dominance that occurs in humans is that of blood type.

ABO blood groups are controlled by the gene I. The plasma membrane of the red blood cells has sugar polymer that produced from its surface and the kind of sugar is controlled by the gene. The gene I has three alleles I^A, I^B and i. the elements A and B produce a slightly different form of sugar while allele i doesn't produce any sugar because humans are diploid

organism. Each person possesses any two of the three I genes allele. I^A and I^B are completely dominant over i . In other words when I and i are present only I^A expresses because i does not produce any sugar. Then I^B and I are present I^B expresses but when I^A and I^B are present together they both express their own type of sugar this is because of Co-dominance. Hence red blood cells have both A and B types of sugars. Since there are three different alleles, there are six different combinations of these three alleles that are possible a total of six different genotypes of the human ABO blood.

Genetic Basis of Blood Groups in Human Population			
Allele from Parent 1	Allele from Parent 2	Genotype of Offsprings	Blood types of Offsprings
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	$I^A i$	A
I^B	I^A	$I^A I^B$	AB
I^B	i	$I^B i$	B
i	i	ii	O

Table showing the genetic basis of blood groups in human population.
